

Analysis of resonant coupled coils in the design of radio frequency transcutaneous links

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Abstract—A theory of coupled resonant coils has been developed which makes possible the design of radio frequency transcutaneous links of simultaneously high overall efficiency and good displacement tolerance while keeping circuitry simple (particularly in the implanted receiver). Series-tuned transmitter coils were used, obtaining high efficiency. In the first example a stimulator which has excellent displacement tolerance because it works at critical coupling is designed. The second example shows how the theory was used when the voltage in the implant was regulated. The design process involves ad hoc compromises between disparate quantities (e.g. efficiency against voltage transfer ratio), rendering a standard design procedure unsuitable. Fortunately, the derived theoretical formulae are simple enough for every design to be considered from fundamentals, based on coil parameters. Extensive coil loss data are presented here for the frequency band 0.2 to 20 MHz.

Keywords—Coil displacement tolerance, Coupled coils, Critical coupling, High efficiency, Implant, Series-tuned transmitter, Transcutaneous radio link

1 Introduction

MAGNETIC coupling between coils at radio frequency is often used for transmitting power, data or both to implanted devices. No general treatment of the design of such coils has been published, perhaps because the specifications imposed are so diverse that entirely different designs result.

Specifications may include requirements concerning:

- (a) overall efficiency of power transfer to the load
- (b) voltage transfer ratio
- (c) phase angle between voltages across the transmitter and receiver coils
- (d) output impedance of the receiver
- (e) displacement tolerance (i.e. the variation in (a)–(d) due to anticipated changes in the coils' relative position)
- (f) size and bulk of the coils
- (g) bandwidth.

The most common requirements need equations relating efficiency and voltage transfer function to coupling coefficient between the coils k ; the displacement tolerance in secondary voltage depends on the rate of change of voltage transfer function with k .

FORSTER (1981) showed that shunt tuning the

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receiver of a pair of previously untuned coupled coils improves displacement tolerance and efficiency. Also, changing the ratio α of secondary reactance to load resistance allowed the efficiency to be maximised for every value of k . A similar result was produced by SCHUDER *et al.* (1961) for a series-tuned receiver coil. From formulae in each of these papers, it can be shown that the maximum obtainable efficiency is approximately $kQ/(kQ+2)$ for $(kQ)^2 \gg 1$ where Q is assumed to be the quality factor for both coils. If, for instance, $Q = 50$, the maximal efficiency will be 71% when $k = 0.1$ with a tuned receiver and an untuned transmitter. However, Forster's equations show that, at this peak in efficiency, the rate of change of secondary voltage is considerable, causing poor displacement tolerance. At high working values of k , such as in Forster's power link which, using ferrite flux guides, had $k = 0.4$, α can be altered to reduce this rate of change without reducing the efficiency much. On the other hand, at the low values of coupling coefficient usually obtained with air-cored coils (0.01–0.15), changing α enough to significantly improve the displacement tolerance has a devastating effect on efficiency. Thus, the disadvantages of this configuration, in which the transmitter coil is untuned, are

- (i) the transmitter circuit has to drive the inductance of the primary coil
- (ii) it is impossible to get good displacement tolerance and efficiency simultaneously at low k

(iii) the untuned reactance in the primary circuit makes the design formulae more difficult.

For these reasons, we used tuned transmitter coils and would only consider not tuning if efficiency was of paramount importance.

It is widely known that with two coupled tuned circuits, a maximum in secondary voltage will occur at some value of coupling coefficient k_{crit} . By corollary, near this maximum, the secondary voltage will be least sensitive to change in k . It follows that operating coils close to $k = k_{crit}$ will make the output voltage tolerant of coil displacement. As the maximum power reaches the receiver at this peak, the efficiency will then be 50% (maximum power transfer condition) although there will be further losses in the receiver coil. If these losses are small, operating at critical coupling should yield reasonable efficiency and good displacement tolerance simultaneously.

The data link designed by FORSTER (1981) worked at critical coupling but his design formulae assume that his receiver was 'tuned to resonance', i.e. $\omega^2 = 1/LC$, from which it follows that the impedance coupled to the transmitter circuit is only real at one value of k . The following Section will show that, at a different frequency, the coupled impedance can remain real for all k . Using this frequency, the simple formulae describing the important properties of the coupled coils are derived.

In Sections 3 and 4, two applications are given, in each case, the theory is used to find an optimal design.

2 Theory

2.1 Preface

The transmitter coil could be shunt or series tuned. To get low k_{crit} with a shunt-tuned coil, the output stage needs to be a good approximation to a current source. As batteries are the usual power supply in portable implant drivers, and these are constant voltage, the output stage would have to produce a constant current from a constant voltage which would be wasteful. Although it is possible to feed a tap on the tank circuit (shunt tuned) from a voltage source, the analysis would be more difficult. The easiest efficient practical circuit to analyse is the combination of series-tuned coil and voltage source.

2.2 Analytical circuit

The coupled circuits are shown in Fig. 1. On the left, a voltage source V_1 of angular frequency ω and with output resistance R_0 is connected to the transmitter coil via a tuning capacitor C_1 . The reactances of the coil and the capacitor add to zero at series resonance but the loss resistance of the coil R_s remains. The transmitter coil is coupled to the receiver by a mutual inductance M at every coil position. The receiver coil L_2 is tuned by C_2 , which is connected in parallel. It is convenient if the loss resistance of this coil also appears

in parallel and this is called R_p . The r.f. waveform appearing across the tuned circuit in the receiver is rectified by the diode D which is connected to the load resistor R_L . The voltage across R_L is smoothed by a large capacitor C_L .

To analyse these coupled circuits, a linear equivalent to D , C_L and R_L is needed. Following KO *et al.* (1977), we may assume that these components act as a perfect peak rectifier so that the peak value of voltage across C_2 appears as a direct voltage across R_L . If the same power is dissipated in R_L as would be dissipated in the equivalent resistor R_{eq} placed directly in parallel with C_2 and L_2 then

$$R_{eq} = \frac{R_L}{2} \quad (1)$$

The circuit of Fig. 1 is redrawn in Fig. 2 with the loss resistors R_s and R_p shown as if they were separate components. For brevity, call

$$R_1 = R_0 + R_s \quad (2)$$

$$R_2 = \frac{R_p R_{eq}}{R_p + R_{eq}} = \frac{0.5 R_p R_L}{R_p + 0.5 R_L} \quad (3)$$

It is convenient also to redraw as if the self and mutual inductances were separate so that the voltage induced in the receiver coil V_i can be shown (Fig. 3).

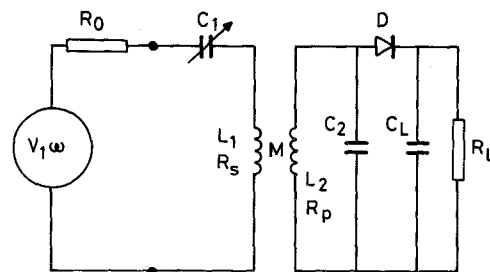


Fig. 1 Series-tuned transmitter coil magnetically coupled to the shunt-tuned receiver coil with rectifier and load

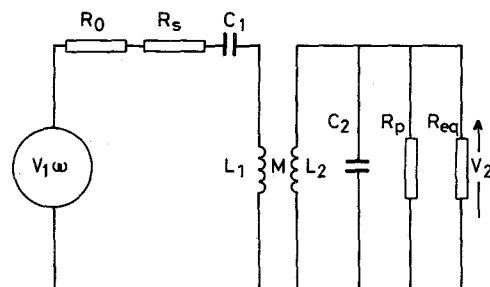


Fig. 2 Same circuit as Fig. 1 but showing the coil losses and a steady-state a.c. equivalent of the actual load

2.3 Tuning

The impedance looking into the receiver from the mutual inductance Z_2

$$= \frac{V_i}{I_2} = \frac{R_2}{(1 + \omega^2 C_2^2 R_2^2)} + j \left[\omega L_2 - \frac{\omega C_2 R_2^2}{(1 + \omega^2 C_2^2 R_2^2)} \right] \quad (4)$$

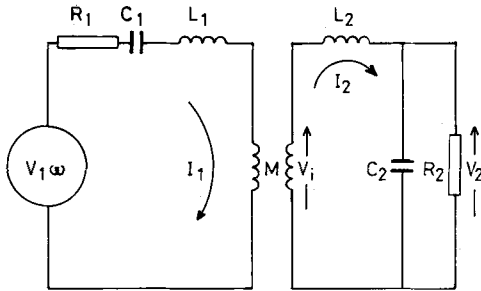


Fig. 3 Circuit in its final analytical form showing the self and mutual inductances separately

This impedance will appear in the transmitter circuit as series impedance $\omega^2 M^2 / Z_2$ (TERMAN, 1943). When the coils are far apart $M = 0$ and the voltage source sees only R_1 at series resonance. When M is not zero, the impedance in the transmitter circuit will be $R_1 + \omega^2 M^2 / Z_2$ and therefore if Z_2 is real, the impedance in the transmitter circuit will be real for all M . The transmitter tuning is then not altered by the presence of the receiver. To make Z_2 real, the imaginary part of eqn. 4 must be zero.

$$\omega^2 = \frac{1}{L_2 C_2} - \left(\frac{1}{C_2 R_2} \right)^2 \quad \text{call this } \omega_0^2 \quad (5)$$

At $\omega = \omega_0$

$$Z_2 = \frac{V_i}{I_2} = \frac{L_2}{C_2 R_2} \quad (6)$$

and

$$\frac{V_1}{I_1} = R_1 + \frac{\omega^2 M^2}{(L_2 / C_2 R_2)} + j \left[\omega L_1 - \frac{1}{\omega C_1} \right] \quad (7)$$

At series resonance in the transmitter and substituting $M^2 = k^2 L_1 L_2$.

$$\frac{V_1}{I_1} = R_1 + k^2 \left(\frac{C_2}{C_1} \right) R_2 \quad (8)$$

Note that in this analysis, the value of C_2 is not

adjusted to maximise V_2 and, indeed, this cannot usually be done with implants. In this case, the frequency of the transmitter must be adjusted to the ω_0 value of the receiver which has been shown to be that at which the impedance appearing in the transmitter circuit is real for all values of coupling coefficient k . The adjustment is complicated because, for every value of ω , C_1 must be changed to maintain series resonance at the transmitter.

2.4 Phase

In the receiver

$$\frac{V_i}{V_2} = (1 - \omega^2 L_2 C_2) + \frac{j \omega L_2}{R_2} \quad (9)$$

when $\omega = \omega_0$

$$\angle \frac{V_i}{V_2} = \arctan \left(\frac{C_2 R_2^2}{L_2} - 1 \right)^{\frac{1}{2}} \quad (10)$$

As V_i leads I_1 by 90° and I_1 is in phase with V_1 for all k , V_2 leads V_1 by a constant angle:

$$\angle \frac{V_1}{V_2} = \arctan \left(\frac{C_2 R_2^2}{L_2} - 1 \right)^{-\frac{1}{2}} \quad \text{at } \omega = \omega_0 \quad (11)$$

2.5 Voltage transfer function

Knowing that the power lost in the coupled resistance term of eqn. 8 is actually dissipated in R_2 , it follows that at $\omega = \omega_0$ the voltage transfer ratio

$\left(A = \left| \frac{V_2}{V_1} \right| \right)$ is given by:

$$A^{-1} = \left(\frac{C_1 R_1}{C_2 R_2} \frac{1}{k} + k \right) \sqrt{\left(\frac{C_2}{C_1} \right)} \quad (12)$$

A has a maximum at a value of $k = k_{crit}$ where

$$k_{crit}^2 = \frac{C_1 R_1}{C_2 R_2} \quad (13)$$

Thus eqn. 12 may be written

$$A = \frac{\sqrt{\left(\frac{R_2}{R_1} \right)}}{\left(\frac{k_{crit}}{k} + \frac{k}{k_{crit}} \right)} \quad (14)$$

And so at critical coupling

$$A_{crit} = \frac{1}{2} \sqrt{\left(\frac{R_2}{R_1}\right)} \quad (15)$$

To find at what values of k , call them k_m , A has fallen by a fraction m below its peak value at k_{crit} , it can readily be shown from eqn. 14 that

$$\frac{k_m}{k_{crit}} = \frac{1 \pm \sqrt{(2m - m^2)}}{(1 - m)} \quad (16)$$

Table 1 shows this ratio for some useful values of m .

2.6 Output resistance

From eqns. 3 and 15 it follows that, at critical coupling, the output resistance of the receiver for small changes in R_{eq} will be

$$R_{out_{crit}} \simeq \frac{1}{\left(\frac{2}{R_p} + \frac{1}{R_{eq}}\right)} \quad (17)$$

2.7 Efficiency

Before finding the theoretical efficiency, it is necessary to define exactly what is to be found. Figs. 1, 2 and 3 show that the two dissipative components of the theory R_1 and R_2 each have two parts (R_0 and R_s ,

R_p and R_{eq} , respectively) and that R_{eq} itself represents power dissipation in the load and in the rectifier of Fig. 1. Furthermore, a real transmitter will not have an ideal voltage source and power lost elsewhere, such as in an oscillator which drives the output stage, must be accounted for. Table 2 represents all these power flows. The overall efficiency should be

$$\frac{P_G}{P_A + P_B + P_C + P_D + P_E + P_F + P_G}$$

However, P_A and P_B depend on circuit design, also P_F is usually small compared with P_G and so it is justifiable convenience to suppose that all power dissipated in R_{eq} (i.e. $P_F + P_G$) is useful. For the circuit of Fig. 1, it is convenient to define the receiver efficiency

$$\eta_R = \frac{P_F + P_G}{P_E + P_F + P_G} = \frac{R_p}{R_p + R_{eq}} \quad (18)$$

the coupling efficiency

$$\eta_C = \frac{P_E + P_F + P_G}{P_C + P_D + P_E + P_F + P_G} \quad (19)$$

and the system efficiency

$$\eta_S = \eta_R \times \eta_C = \frac{P_F + P_G}{P_C + P_D + P_E + P_F + P_G} \quad (20)$$

Table 1. Ratio k_m/k_{crit} for some useful values of m

m	% down	$\frac{k_m}{k_{crit}}$ upper	$\frac{k_m}{k_{crit}}$ lower	Ratio of two values of k_m
0	0	1.000	1.000	1.00
0.05	5	1.381	0.724	1.91
0.1	10	1.595	0.627	2.54
0.2	20	2.000	0.500	4.00
0.3	30	2.449	0.408	6.00

Table 2. Total power consumption

Other power losses		Power dissipated in circuits of Fig. 1			
Oscillator etc.	Dissipation in output stage not accounted for by R_0 , e.g. offsets, overlap current (see Section 3.6)	Transmitter side (R_1)		Receiver side (R_2)	
		Power in R_0	Power in R_s	Power in R_p	Power in R_{eq}
					* Useful power
P_A	P_B	P_C	P_D	P_E	P_F P_G

* Rectifier losses.

From eqn. 8 the coupling efficiency may be written

$$\eta_c = \frac{k^2 \left(\frac{C_2}{C_1} \right) R_2}{R_1 + k^2 \left(\frac{C_2}{C_1} \right) R_2} = \frac{k^2}{k^2 + k_{crit}^2} \quad (21)$$

η_c rises from 0, at $k = 0$, to 0.5 at $k = k_{crit}$ and reaches $\frac{1}{1 + k_{crit}^2}$ (i.e. approximately unity for small k_{crit}) for $k = 1$. The normalised functions of voltage transfer function and coupling efficiency are shown in Fig. 4.

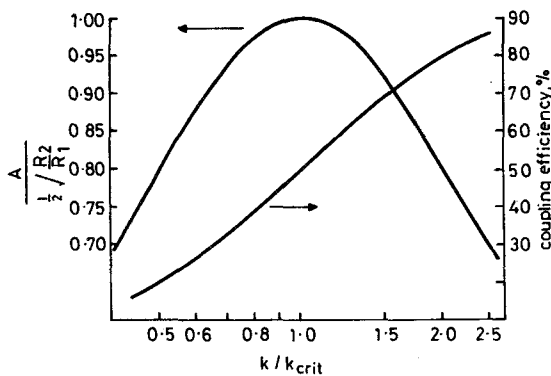


Fig. 4 Normalised functions for A (eqn. 14) and η_c (eqn. 21)

2.8 Bandwidth

At frequencies other than ω_0 , the algebra of the system is more difficult. Fortunately, an approximate solution has been found by AIKEN (1937) for narrowband filters. His results have been summarised by VALLEY and WALLMAN (1948) and their terminology is used. Although they have an equation describing the bandpass characteristic when the coils are tuned apart (as in this case), a simple approximation is to calculate for the case when $L_1 C_1 = L_2 C_2$ and remember that slight stagger tuning is bound to broaden the band.

VALLEY and WALLMAN (1948) define the variables b and v :

$$v = \frac{1}{k_{crit}} \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) \quad (22)$$

$$b = \left(\frac{Q_1}{Q_2} + \frac{Q_2}{Q_1} \right) \quad (23)$$

where

$$Q_1 = \frac{1}{\omega_0 C_1 R_1} \quad \text{and} \quad Q_2 = \omega_0 C_2 R_2.$$

From their second equation on p.203, at critical coupling

$$v_{3dB} = \left[\left[\left(\frac{b}{2} - 1 \right)^2 + 4 \right]^{\frac{1}{2}} - \left(\frac{b}{2} - 1 \right) \right]^{\frac{1}{2}} \quad (24)$$

Thus, as the Q ratio changes from one to infinity, v_{3dB} falls from $\sqrt{2}$ to 0, reaching 0.68 when the ratio is 10. For low values of critical coupling the fractional bandwidth is small and eqn. 24 gives

$$B \simeq v_{3dB} k_{crit} \quad (25)$$

The bandwidth increases with coupling coefficient. This is shown by TERMAN (1943).

3 First example: A design to give good coil displacement tolerance

3.1 Specification

The requirement here is a common one; to design, within the constraint that neither the implant nor the external coil is too large, so that the stimulating voltage is as tolerant of coil position as possible.

As maximal coupling coefficient and absolute displacement tolerance would be achieved with the largest receiver coil, this was chosen from surgical considerations to be 45 mm diameter. The anticipated coil spacing was 25 mm. To achieve acceptable device thickness, the implant coil was to be no longer than 2.5 mm and the transmitter coil no longer than 3.5 mm. *In vitro* tests with the 100 μ s pulses used indicated that the electrodes and associated cable would present a largely resistive impedance of 200 Ω (for much longer pulses electrode capacitance would effectively increase this impedance). The largest stimulation voltage would be 15 V and the power supply to the transmitter circuit should be the same, so that this could run in practice from two 8.4 V batteries in series with some regulation.

A carrier frequency of 4.0 MHz was to be used (see Section 3.8).

3.2 Design method

The actual coupling coefficient with typical coils at typical working spacings is small and it is not always possible, by design, to reach such small values for critical coupling. The receiver voltage will be most stable if k changes slowly with displacement in the operating region. The first stage in the design was therefore to choose the transmitter coil diameter so that k is as large as possible and also changes slowly. This might entail a compromise.

Having both coil diameters and spacing, a value of k in the working position follows, which should be the value of k_{crit} . The next step was the choice of coils (number of turns, wire diameter, single or double layer). As all the coils considered were short and cylindrical, the coupling coefficient was assumed to be

independent of these coil details, depending only on diameter. To set k_{crit} to the desired value, C_1 , R_1 , C_2 and R_2 were needed (eqn. 13). R_1 and R_2 include, respectively, the output resistance R_0 and the load resistance R_{eq} but R_s and R_p are coil properties which are frequently dependent (Fig. 2). Sets of coils were wound with the chosen diameters with differing wire size and winding. Measurements were made of inductance, loss resistance (R_s or R_p) and self capacitance.

With the transmitter circuit used here, the output resistance depends mainly on the size of the pair of complementary power m.o.s.f.e.t. transistors. Decreasing R_0 will reduce R_1 and hence reduce k_{crit} as desired. However, larger power devices have greater input capacitance which must be driven by the previous stage, consuming power, and lowering the overall efficiency. As the advantageous reduction in k_{crit} will diminish, for decreasing R_0 , as R_0 becomes small compared with R_s , a balance must be struck between low k_{crit} and efficiency. Our approach was to design an oscillator and output stage for the transmitter which drew a small current compared with the load (i.e. the series-tuned circuit coupled to the receiver) and measure the value of R_0 achieved. This value was then added to the transmitter coil resistance R_s , to give R_1 . The values of k_{crit} and A_{crit} could then be estimated for all likely pairs of coils.

From these data, the most suitable pair was chosen. k_{crit} must be equal to, or just less than, the working value of k (R_0 can always be adjusted with a small 'padding' resistor to get the exact value of k_{crit} required), and as A_{crit} will determine the battery voltage needed for the specified maximal stimulating voltage, this will affect the choice. The receiver efficiency (η_R , eqn. 18) must not be ignored. Decreasing the number of receiver coil turns and hence parallel loss resistance will reduce η_R but decrease k_{crit} by increasing $C_2 R_2$. This is yet another compromise.

To summarise, the choice of coils in this example requires three balances to be struck to achieve the required properties of the system:

- (a) for the transmitter coil diameter: higher working k against displacement tolerance
- (b) driver stage output resistance R_0 : lower k_{crit} against losses in the previous stage
- (c) receiver coil loss resistance R_p : lower k_{crit} against losses in the receiver.

The overriding aim is to get k_{crit} at the working value of k with the best compromise between efficiency and convenient voltage transfer ratio, within the overall size constraint. An iterative procedure involving a number of possibilities may be necessary.

3.3 Choice of transmitter coil diameter

To do this, the variation in coupling coefficient must be found with coil diameter and with small

displacements for different coil diameters.

k was measured between pairs of a set of coils, all 2 mm long but of various diameters, at coaxial spacings of 25 mm. The experiment involved measuring the resonant frequency of a fixed capacitor and the total inductance of both coils of each pair when connected in series, using a crystal-standard frequency counter. As the mutual inductance can be positive or negative, depending on the connection, the total inductance is $L_1 + L_2 \pm 2M$, and so two frequencies were measured from which k could be deduced. However, this involves finding the small difference between two large numbers and is not very accurate. The results, in Fig. 5, show that as the transmitter coil diameter varies, k has a broad maximum at about 63 mm.

KO *et al.* (1979) have given a formula for finding the position of this maximum, produced by differentiating a theoretical equation given in TERMAN (1943). The formula is

$$d_1^2 = d_2^2 + 4D^2 \quad (26)$$

where d_1 and d_2 are the coil diameters and D is the spacing. This is not strictly applicable in this case because the formula applies to coils of constant shape factor (length/diameter) which these are not. However, as the length of all these coils is short compared with the spacing, it would be surprising if this caused much error and, indeed, the predicted peak occurs at $d_1 = 67$ mm; as marked by the arrow on Fig. 5.

The numerically computed curves of FLACK *et al.*

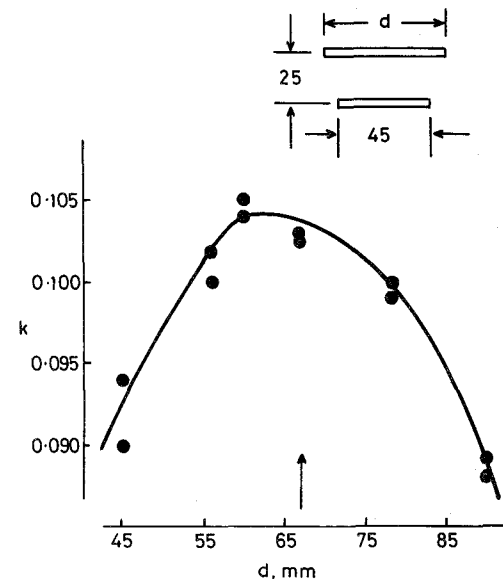


Fig. 5 Measured variation in coupling coefficient with transmitter coil diameter. All coils were 2 mm long

(1971) were used for considering the variation of k with displacement. Some experimental justification for trusting these curves is given below. These curves apply to pairs of coils with diameter ratios of 1.0, 0.75, 0.5 and 0.25 which, with a fixed receiver coil diameter of 45 mm, gives possible transmitter coils of 45, 60, 90 and 180 mm. For the first three sizes, the displacements needed to cause a change in coupling coefficient of $\times 1.381$ or $\times 0.724$ (i.e. the 5% down values of A , see Table 1) were estimated in Table 3. This shows that the displacement tolerance improves substantially as the transmitter coil diameter rises from 60 to 90 mm. Also, from Fig. 5, the value of k is only about 15% below its maximum with a 90 mm coil. From these considerations a good compromise would be a transmitter coil of 80 or 90 mm diameter. Unfortunately, this was too large for the patient to carry, so 60 mm was chosen, giving a higher k but poorer displacement tolerance.

To verify the FLACK results for our chosen pair of coils, k contours were extracted from their results and plotted as shown in Fig. 6. As their results give mutual inductance factor and not k itself, we have normalised using several measured axial k values. The stars on Fig. 6 show measured positions for the given k value. As all the stars are consistent with the computed curves to within 0.5 mm, the curves seem at least accurate enough for practical purposes.

3.4 Coil measurements

The loss resistance of coils is frequency dependent and although this example deals with one frequency,

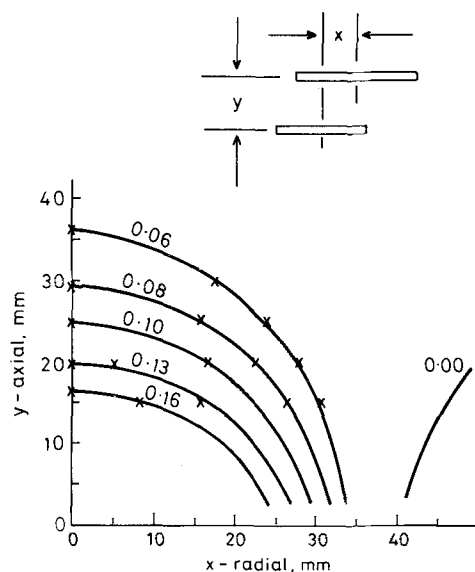


Fig. 6 Contours of constant k taken from FLACK et al. (1971) for the coil configuration shown. The coils are 45 and 60 mm diameter. The stars mark our measured positions of these values of coupling coefficient

Table 3. Displacement

Transmitter coil diameter, mm	Displacements from a coaxial spacing of 25 mm		
	Radial, mm	Axial increased spacing, mm	Axial decreased spacing, mm
45	17	4.5	4.5
60	19	6.5	6.0
90	32	9.5	10

Table 4. Receiver coils

Number of turns	Wire diameter, mm	Winding single or double layer	L , μH	C_p , pF
1	0.9	S	0.131	—
1	0.5	S	0.148	—
1	0.375	S	0.159	—
2	0.5	S	0.48	4.7
2	0.375	S	0.514	4.0
2	0.25	S	0.569	3.6
4	0.5	S	1.64	5.1
4	0.375	S	1.76	5.6
4	0.25	S	1.99	5.1
4	0.15	S	2.20	5.1
8	0.375	D	6.64	12.4
8	0.375	S	6.13	3.8
8	0.25	S	6.76	3.3
8	0.15	S	7.67	3.5
8	0.08	S	8.69	3.7
16	0.25	D	25.7	20.3
16	0.15	D	28.4	21.7
16	0.08	S	30.2	3.5

Table 5. Transmitter coils

Number of turns	Wire diameter, mm	Winding single or double layer	L , μH	C_p , pF
2	0.5	S	0.681	5.9
4	0.5	S	2.34	7.0
4	0.375	S	2.53	7.0
6	0.5	S	4.84	5.2
8	0.375	S	8.75	5.0
8	0.25	S	9.69	4.9
12	0.25	D	22.1	20.7
12	0.25	S	19.9	4.7
16	0.25	D	37.6	25.8
16	0.15	D	42.1	23.8
32	0.15	D	147	46

loss resistance was measured over the band usually employed in radio links, the two decades from 0.2 to 20 MHz. The length of the coils was limited by the acceptable thickness of the implant coil (<2.5 mm) and the external 'driver', containing the transmitter coil (<3.5 mm). The coils tested have up to 32 turns and windings were split into double layers, 'layer wound' (TERMAN, 1943, section 2.2) when these limits would otherwise be exceeded. The range of wire diameter used was 0.08 to 0.5 mm with the wire gauge for any number of turns selected so that the volume of copper remained approximately the same. All the wires were single-strand, polyurethane coated and the

coil was impregnated with silicone rubber adhesive (Dow Corning 734), our usual encapsulant.

Each coil was tuned, in turn, by several different standard capacitors, connected in parallel. A current from an r.f. sinusoidal source was applied via a resistor. The voltages across the tuned circuit, and the tuned circuit plus resistor, were monitored with an oscilloscope while the frequency was adjusted to bring the waveforms in phase ($\omega^2 = 1/LC$). The relative amplitudes of these voltages and the value of the fixed resistor then gave the 'parallel loss resistance' for the resonant circuit. It was found that capacitors with the lowest loss had to be used if all this resistance was to be

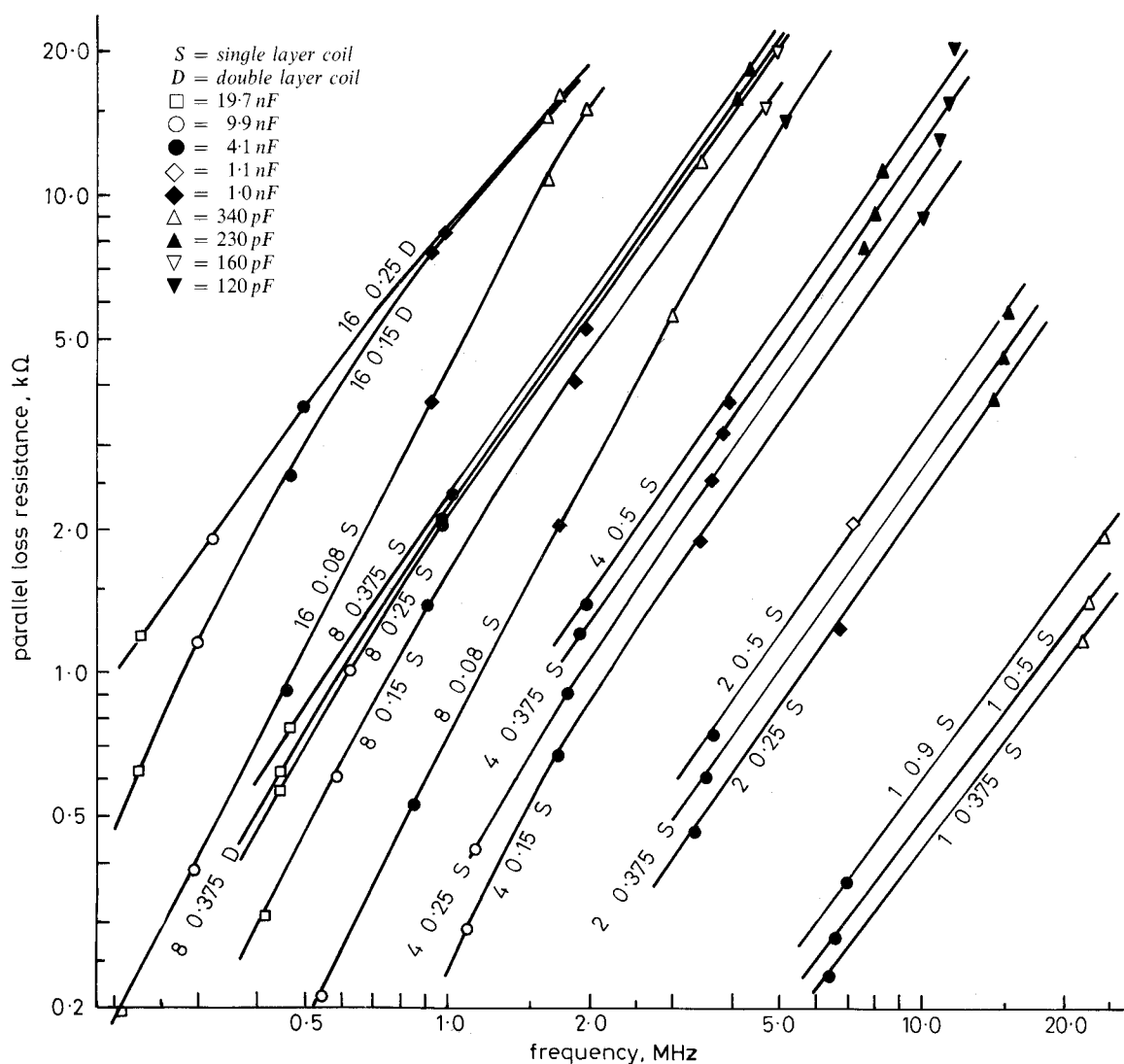


Fig. 7 The measured variation of parallel loss resistance with frequency for 45 mm diameter, cylindrical, close-wound coils. The curve for each coil is marked: number of turns: wire diameter: single or double layer

fairly attributed to the coil. With fixed capacitors that were not large compared with the input capacitance of the oscilloscope probes, the probe loss had to be taken into account and the loss resistance figure corrected.

For the 60 mm transmitter coils, the series loss resistance is required and this was calculated from the measured figure by the approximation

$$R_s = \omega^2 L^2 / R_p.$$

A 'grid dip meter' was used to measure the self-resonant frequency of each coil in isolation. Knowing the coil inductance from measurement with a large tuning capacitor, the self capacitance could be calculated (this fictitious lumped component is imagined in parallel with the whole coil).

Measurements from all the coils are collected in Tables 4 and 5 and Figs. 7 and 8.

For all the transmitter coils, the frequencies were calculated at which the self capacitance reaches 5 and 10% ($p = 0.05, 0.10$) of the total tuning capacitance. These frequencies are marked by stars on Fig. 8 and give a rough indication of the upper frequency limit for the coil when series tuned.

Fig. 8 shows that the series resistance of coils rises linearly on a log/log graph, increasing by a factor of about 1.5 with every doubling of frequency. This holds down to low frequencies where the resistance tends to the d.c. value. On the receiver coil curves, this tendency causes decreasing resistance (greater loss) in the coils with finer wire for a given number of turns at given frequency.

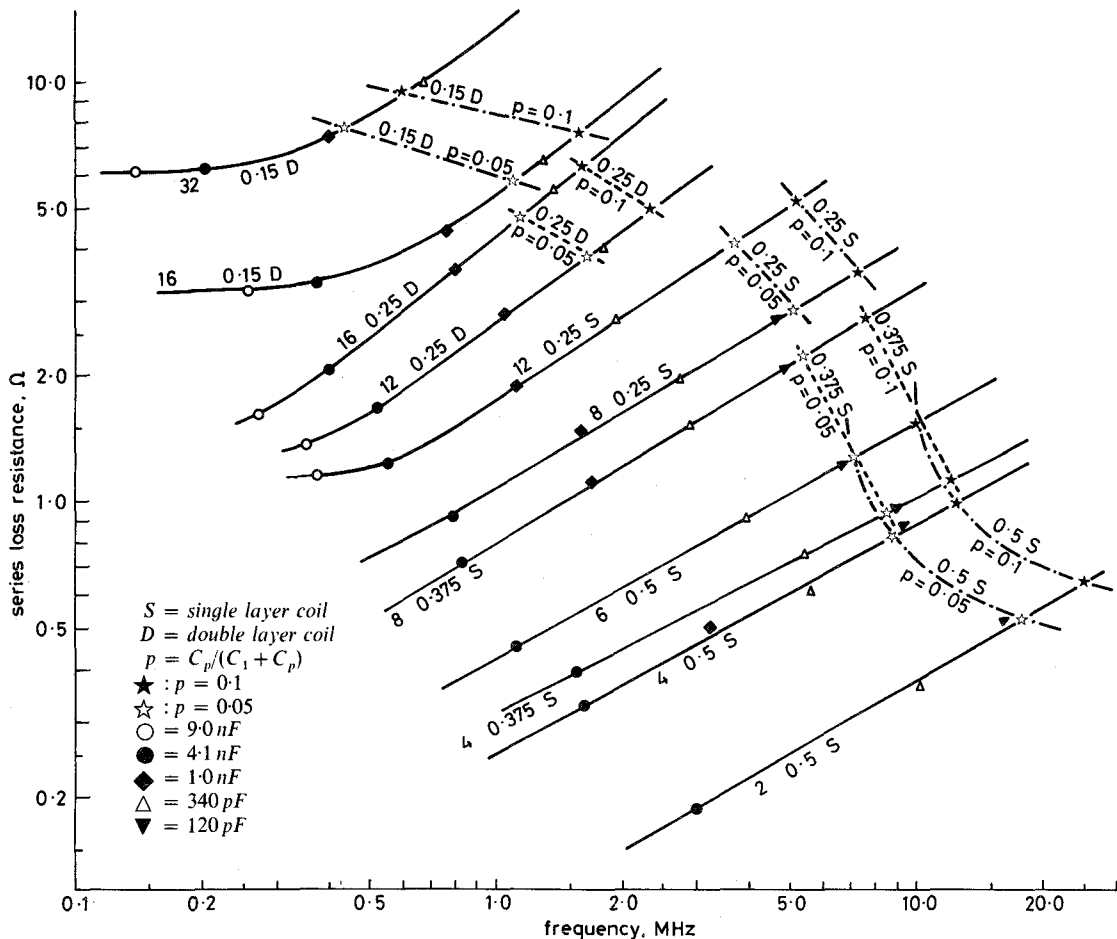


Fig. 8 The measured variation of series loss resistance with frequency for 60 mm diameter, cylindrical, close-wound coils. The curve for each coil is marked: number of turns: wire diameter; single or double layer. The stars show the frequencies at which the self capacitance reaches 5 or 10% of the total tuning capacitance

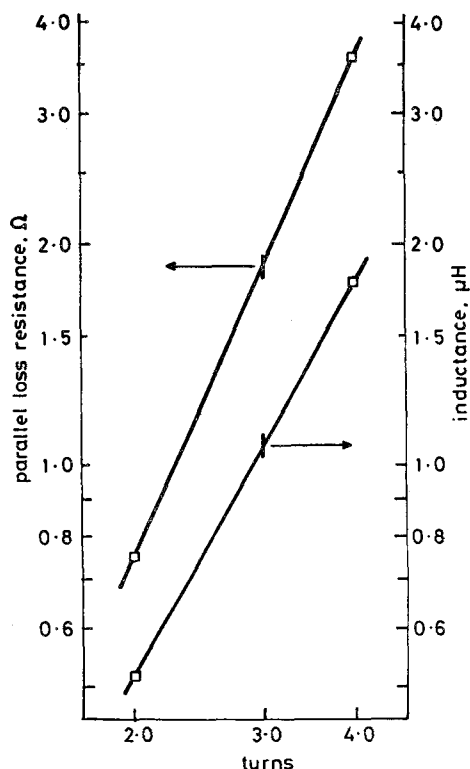


Fig. 9 Log/log graph to find the parallel loss resistance at 4 MHz and inductance of a three-turn coil by interpolating between values for two and four-turn coils, taken from Fig. 7 and Table 4. All three coils have 0.375 mm wire diameter

To check the accuracy of the loss resistance measurements, the R_p values for the four-turn coil of 0.25 mm wire were compared with those computed from a reference (BEATTY and SOWERBY, 1949) over the range 1.1 to 11 MHz. All the points agreed within 10%, the average of the measurements being 4% low.

3.5 Choice of coils

For high receiver efficiency, the parallel loss resistance R_p must be large compared with R_{eq} which is 100 Ω here. If coils are considered for which this is so, the value of R_2 will remain close to 100 Ω. The inductance values of Table 4 show that there will, however, be considerable variation in C_2 at 4 MHz. For high $C_2 R_2$, giving low k_{crit} , the coils with the lowest inductance are needed. Referring to Fig. 7, the two-turn coil of 0.5 mm wire has a resistance of 900 Ω at 4 MHz giving $\eta_R = 90\%$. The four-turn coils, having higher resistances, will be more efficient still, but as the one with 0.5 mm wire is rather long, the 0.375 mm coil was considered. To improve the choice, a three-turn coil was included. The predicted values for inductance and loss resistance were found by interpolation on a log/log graph between the values for two- and four-turn coils of 0.375 mm wire (see Fig. 9). The values derived were 1.06 μH and 1.87 kΩ which were later checked to be within 1% of measured values. For all three possible receiver coils, R_2 was calculated and hence C_2 using eqn. 5, as shown in Table 6.

Similar reasoning exists when selecting the possible transmitter coils. As R_1 is restricted by the output resistance of the driver stage, the product $C_1 R_1$ is minimised by having the maximum number of turns.

Table 6. Calculated values for possible coils

Coil	Transmitter coils			
	12 turns 0.25 mm*	8 turns 0.25 mm	8 turns 0.375 mm	6 turns 0.5 mm
C_1 , pF	79.5	163	181	327
R_1 , Ω	9.45	7.57	6.96	6.02
$C_1 R_1$, ns	0.75	1.23	1.26	1.97

Receiver coils				A_{crit}/k_{crit} for all coil combinations			
coil	C_2 , nF	R_2 , Ω	$C_2 R_2$, ns				
4 turns 0.375 mm	0.64	97	61.6	1.60/0.110	1.79/0.141	1.87/0.143	2.01/0.179
3 turns 0.375 mm	1.36	95	129	1.59/0.076	1.77/0.098	1.85/0.099	1.99/0.124
2 turns 0.5 mm	3.23	90	291	1.54/0.051	1.72/0.065	1.80/0.066	1.93/0.082

There are twelve pairs of coils represented in the Table.

Values for $k_{crit} = \sqrt{\frac{C_1 R_1}{C_2 R_2}}$ and $A_{crit} = \frac{1}{2} \sqrt{\frac{R_2}{R_1}}$ are given for each pair.

* $p > 0.05$ for this transmitter coil.

However, if the self capacitance becomes significant, account has to be taken of its effects and it is easiest to consider coils with $p < 0.05$. (Note that coils with many turns, having only small tuning capacitors, will be significantly detuned by stray capacitance such as that to the patient.) Four transmitter coils are considered, each R_1 being found by adding their resistances R_s at 4 MHz to the measured value of the output resistance of the driver (5.1Ω see Section 3.6); $C_1 = 1/(\omega_0^2 L_1)$. For all the 12 possible coil combinations, values of k_{crit} and A_{crit} were found, using eqns. 13 and 15, and these are shown in Table 6.

As, for this example, k_{crit} needs to be about 0.10, several pairs have a value of k_{crit} low enough. Both the eight-to-three turn pairs have very close values and the transmitter coil with 0.25 mm wire was chosen for compactness.

3.6 Transmitter circuit

The theory given in Section 2 assumed a sinusoidal r.f. voltage source V_1 with a fixed output resistance R_0 as a usable model. The circuit described here (Fig. 10) is actually a square wave source, with the output switched between two saturated states, which gives high efficiency and low R_0 simultaneously, and yet may be approximated by a sinusoidal source (at least in steady state). This is due to the filtering of the series-tuned coil which allows power dissipation from the odd harmonics according to

$$\frac{P_n}{P_1} \approx \left[\frac{2}{n^2 Q_1 (1-k)^2} \right]^2 \quad (27)$$

This is an extremely small fraction even for $n = 3$ for the coils considered in the previous Section. It is therefore a reasonable approximation that a pure sinusoidal current of the fundamental frequency is drawn in phase with the square wave drive. The square wave switches the output between the power supply lines, a voltage swing of V_s ; multiplying by the Fourier coefficient for the first harmonic of a square wave leads to an expected value of $V_1(\text{peak}) = \frac{2V_s}{\pi}$.

In the circuit of Fig. 10, the BC 337 is the master oscillator transistor, oscillating at the resonant frequency of the 28 turns of the toroidal inductor and the 68 pF c.o.g. capacitor plus an 18 pF trimmer. This is adjustable about 4.0 MHz. The complementary m.o.s.f.e.t. devices, connected in common drain configuration, need a single secondary winding to switch them in antiphase. The oscillograph in Fig. 11 shows the gate/source voltage with a peak amplitude of 22 V at a power supply voltage of 15 V. This amount of drive to the output devices causes reasonably rapid transitions at the output shown under load at k_{crit} on the same Figure. Also shown is the voltage across the resonant circuit of the receiver with a 100Ω a.c. load, using the coils chosen in the previous Section.

With a capacitor across the toroidal transformer secondary to represent the sum of the two gate capacitances, the master oscillator alone drew 26 mA from a 15 V supply. With the output m.o.s.f.e.t.s in circuit but without the transmitter coil, the current rose substantially. This was mostly due to 'overlap current' flowing through both devices when they were conducting simultaneously, presumably due to their different switching speeds. This current was minimised

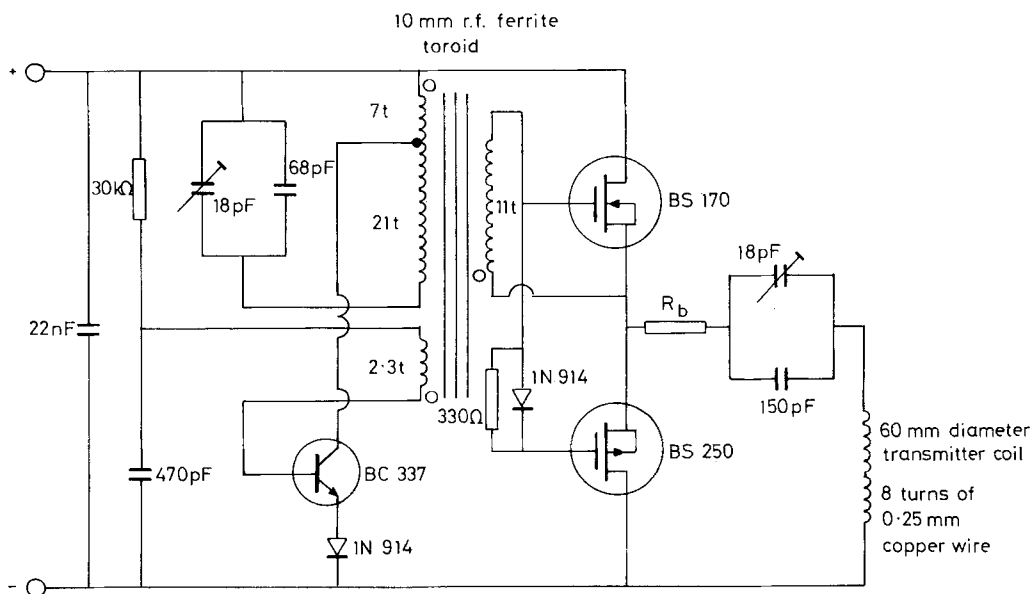


Fig. 10 Transmitter circuit

by the diode and $330\ \Omega$ resistor shown in the circuit diagram. The overlap current was then 37 mA. These two currents, totalling 63 mA, were subtracted from the total current drawn when the transmitter coil was connected to give a 'load current' I_s .

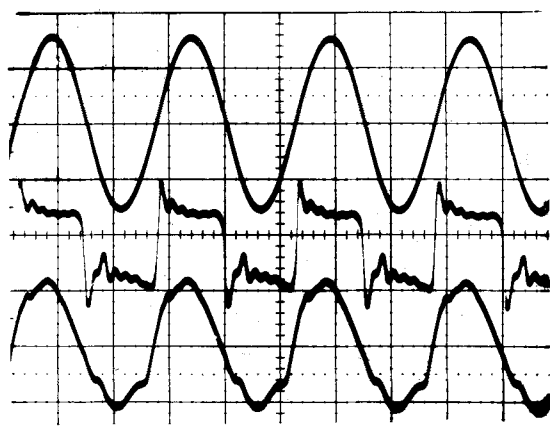


Fig. 11 Oscilloscope taken when the transmitter circuit is operating at critical coupling. The horizontal scale is 100 ns/division. Lower trace; gate-source voltage for one output m.o.s.f.e.t. (20 V/division); middle trace, voltage at the two sources (10 V/division); and the upper trace is the voltage across the receiver coil with a 100 Ω a.c. load (10 V/division)

The output resistance of the transmitter circuit was measured at 4.0 MHz ($V_s = 15$) as $5.1\ \Omega$ (the value used in Section 3.5). The 'padding' resistor R_b was made $0.82\ \Omega$ to yield k_{crit} at a coaxial spacing of 25 mm.

3.7 Performance of the system

The design, now complete, was tested to see if it performed as expected. For accurate numerical comparison between the measured and theoretical values, account should be taken of the effect of the self capacitance of the transmitter coil (see the Appendix). In this case $p = 0.03$ (Table 5), so R_0 (including R_b) was reduced by 6%, i.e. to 0.94 ($5.1 + 0.82$) = $5.57\ \Omega$. This was added to the transmitter coil resistance ($2.47\ \Omega$, Fig. 8) to give $R_1 = 8.03\ \Omega$. For the receiver, the exact value of R_p is $1.87\ \text{k}\Omega$ (Fig. 9), giving R_2 as $94.9\ \Omega$ if R_{eq} is $100\ \Omega$. C_1 and C_2 are as shown in Table 6. These values gave exact theoretical values for k_{crit} (eqn. 13) of 0.101 and A_{crit} (eqn. 15) of 1.72 . This k_{crit} value is close (3%) to the measured value of k at 25 mm spacing shown in Fig. 5. As previously explained, the transmitter power supply voltage must be multiplied by a Fourier coefficients of $\frac{2}{\pi}$ to give the peak value of

V_1 . This is then multiplied by A_{crit} and $(1 - p)$, when the self-capacitance makes this significant, to give V_{2crit} (peak).

Thus, with a 15 V power supply, the expected peak value of

$$V_{2crit} = 15 \times 2/\pi \times 0.97 \times 1.72 = 15.9.$$

The peak voltage measured at critical coupling across the receiver coil was 15.4 V. According to the considerations on efficiency given in Section 2.7, of 100 units of power entering the transmitter tuned circuit, 50% will be dissipated in the receiver at critical coupling (η_c) and a fraction η_R of that will be in the load (R_{eq}). In this case, η_R should have been 94.9% (eqn. 20) and so $\eta_R \times \eta_c = \eta_s = 47.5\%$. Having placed the coils as close as possible to critical coupling, η_s was actually measured as 47.7% (the input power being measured as $I_s \times V_s$). Measuring the power dissipated in the load R_L after rectification (Table 2), the efficiency had fallen to 44.7% due to losses in the diode (1N4148). These figures were measured with a 15 V power supply. If the dissipations in the master oscillator and the output stage due to 'overlap current' were accounted for, an additional 38% was added to transmitter consumption. This means that the overall efficiency (Section 2.7), from battery to load resistor, is $44.7 \times 100/138 = 32.4\%$, a very satisfactory value.

The position tolerance of the system was found by

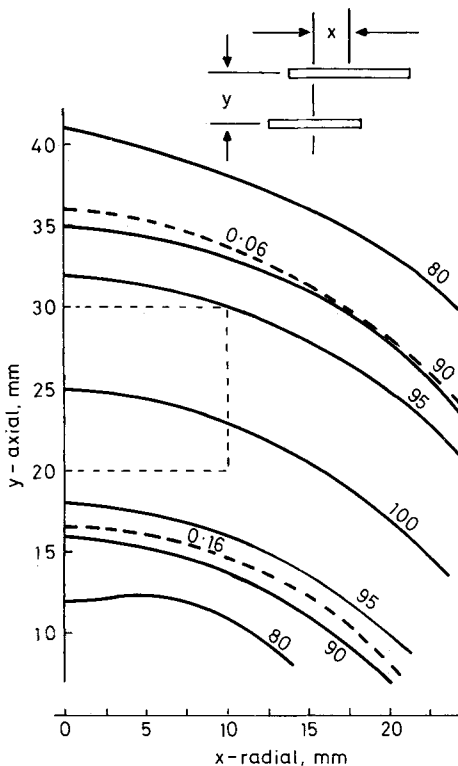


Fig. 12 Measured contours of constant receiver voltage, shown as a percentage of maximum. The dashed curves are two k contours from Fig. 6. The box indicates a possible operating region

plotting contours of constant output voltage as a function of coil position; see Fig. 12. Voltages are given as a percentage of the peak value at critical coupling. Also shown here are two of the k contours shown on Fig. 6. The broken square indicates what tolerance has been achieved: axial movements of ± 5 mm and possible simultaneous radial movements of 10 mm will not cause the output voltage to change by more than 5%. The 10% down lines lie along $k = 0.064$ and 0.159 , a ratio of 2.50 apart, which is close to the predicted value of 2.54 (Table 1). Note the agreement (± 1 mm) between the expected values for displacement tolerance given in Table 3 and the 5% down curves in this Figure.

The receiver output impedance at the position of critical coupling should have an output resistance of 90.2Ω (eqn. 17) and was measured as 95Ω . As V_1 is not available for measurement, the phase was observed with respect to the master oscillator. As accurately as could be measured ($\pm 10^\circ$), the angle remained constant for all receiver positions.

The benefit in position tolerance from operating coupled coils near critical coupling has been fully exploited. Improvements could only be obtained with coils larger than is acceptable (Section 3.3). Meanwhile the efficiency is high and the required transmitter power supply voltage is convenient. Nevertheless a new design might be slightly better in terms of the three compromises given at the end of Section 3.2. It seems likely that a slightly smaller receiver coil, which the surgeon would no doubt welcome, could still be critically coupled and be as displacement tolerant. Alternatively, a higher overall efficiency might be achieved if η_R were decreased but a higher impedance output stage substituted decreasing oscillator and overlap current, to maintain k_{crit} , giving a greater efficiency improvement. With this particular design of transmitter circuit, improvement may come with better small m.o.s.f.e.t.s, having lower gate capacitance for the desired 'on' resistance.

3.8 Choice of frequency

It is useful to consider how changing the carrier frequency could affect performance.

Generally, the frequency band that may be used lies between an upper limit imposed by tissue absorption (CHRISTENSEN and DURNEY, 1981) and a lower limit depending on signal bandwidth and possibly the need to avoid radio interference on the medium waveband. With devices like this example, a range of 1.5 to 10 MHz would be possible.

Figs. 7 and 8 can be used to investigate the effect of different frequencies. In the receiver, with a given load resistance, η_R will be maintained by keeping R_p constant, which means moving horizontally across Fig. 7, approximately halving the number of turns for every trebling of frequency. Following the $p = 0.05$ lines on Fig. 8 suggests that R_s will fall about fivefold and the number of turns about threefold over the same

frequency ratio. Neglecting R_0 , these changes imply a slow fall in k_{crit} ($\times 0.65$) and an increase in A_{crit} ($\times 2.0$) with this frequency shift. In fact, R_0 is not negligible and increases as the frequency rises, as does the overlap current and master oscillator current. R_0 may be decreased by using larger transistors but at the expense of further increases in these currents. Again, improvement in k_{crit} is counteracted by falling overall efficiency.

In practice, one approach is to try a frequency at which the power used by the master oscillator and wasted as 'overlap current' in the driver circuit is a reasonable fraction of the power required at the load. This will give a value of R_0 from which to choose coils, as shown in the example. When the performance of this system has been estimated, changing the frequency downwards should favour efficiency, increasing frequency should allow greater working coil spacings.

4 Second example: receiver Zener regulated to provide a constant voltage power supply

4.1 Specification

The receiver is to have a Zener diode connected across its d.c. output to clamp the voltage to 20 V. This regulator is arranged so that a demultiplexer in the implant only operates when 20 V is reached and is then synchronous with an external clock. This is to occur at a coil spacing of 29 mm when the implant 'locks in'. Three milliamps are drawn from the power supply at 20 V. The No. 2 morphognostic coil shape (the 'figure of eight' coil (DONALDSON, 1979)) is to be used as there is a separate data channel using circular coils. A 7.8 MHz series-tuned transmitter is available with this coil shape. The coil of this transmitter is 48 mm diameter and wound cylindrically with 0.375 mm

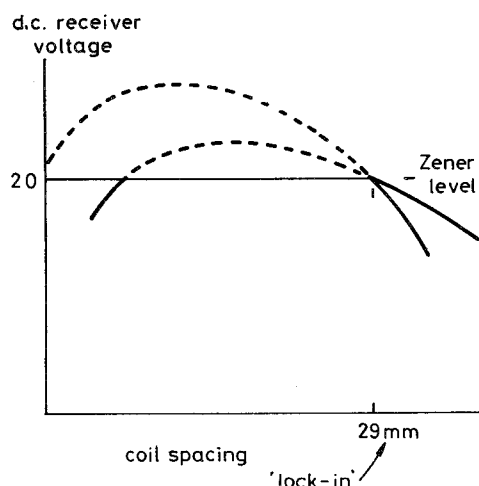


Fig. 13 This sketch shows the d.c. output voltages from two receivers with different coils. In both cases, the 20 V Zener regulator level is reached at a 29 mm coil spacing

copper wire. The tuning capacitor is 140 pF and $p = 0.06$.

The corresponding coil in the implant may be the same diameter but could only use 0.25 mm wire scramble wound to keep the implant thickness down. The problem is to find the number of turns to use in this coil, to minimise the transmitter voltage needed to produce 'lock-in' at 29 mm.

4.2 Use of the theory

With the Zener conducting and the output voltage for the receiver clamped, the theory given in Section 2 is invalid; the efficiency will be lowered by power dissipation in the Zener diode. However, at the point of 'lock-in', the Zener current is negligible, $R_{eq} = 0.5 \times \frac{20}{3} = 3.35 \text{ k}\Omega$, and the theory is applicable. As the load resistance is constant, the requirement to minimise transmitter voltage may be restated as requiring the highest efficiency at 'lock-in'.

Fig. 13 shows a sketch of output voltage against spacing for two receiver coils, the transmitter voltage having been adjusted so that both pass through 20 V at 29 mm spacing. The two curves have different values of k_{crit} due to varying the number of receiver turns. The system efficiency at 'lock-in' can be calculated from eqns. 18, 20 and 21 once the values of critical coupling and loss resistance R_p have been measured for different coils and k is known at 29 mm.

4.3 Measurements

Receiver coils were wound according to the specification. They were tuned to 7.8 MHz and their loss resistances measured. Connected as receivers with 6.7 k Ω loads but without the Zener diode, the spacings of critical coupling were measured. These results are shown in Table 7.

The figures in the last column of the Table are taken from the curve shown in Fig. 14 which was also measured. Note k at 29 mm is 0.017.

4.4 Calculation of efficiencies

Taking R_{eq} as 3.35 k Ω , η_R and η_C can be calculated from these results and their product found for the three coils at 29 mm spacing (Table 8).

Table 7. Spacing of critical coupling

Number of turns	Loss resistance R_p , k Ω	Spacing of critical coupling with given transmitter, mm	k_{crit}
5	9.5	15	0.066
4	6.8	16	0.056
3	4.85	17	0.049

4.5 Results

The Tables show that as the number of turns in the receiver coil falls, k_{crit} falls and hence the coupling efficiency rises as k_{crit} becomes closer to k . However, this is accompanied by a falling loss resistance which lowers η_R . The system efficiency is highest with four turns, although three turns is only slightly inferior.

To confirm the optimal number of turns, the three coils were connected separately to the Zener-regulated circuit and the transmitter voltage adjusted, in each case, to just produce 'lock-in' at 29 mm spacing. The results, shown in Table 9, agree with the optimum predicted by the theory.

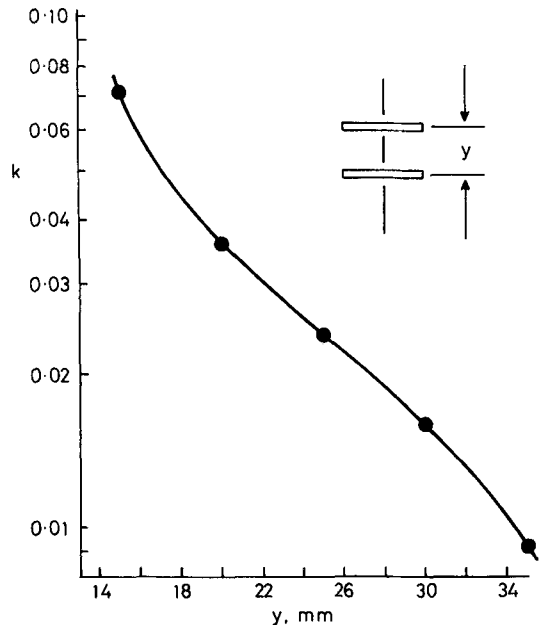


Fig. 14 Coupling coefficient against spacing for a pair of 48 mm diameter No. 2 morphognostic coils which are coaxial and aligned

Table 8. Calculation of efficiencies

Number of turns	$\eta_R = \frac{R_p}{R_p + R_{eq}}$	$\eta_C = \frac{k^2}{k^2 + k_{crit}^2}$	$\eta_S = \eta_R \eta_C$
5	0.74	0.062	0.046
4	0.67	0.084	0.056
3	0.59	0.107	0.054

Table 9. Transmitter voltage at 'lock in' for different numbers of turns

Number of turns	Transmitter voltage at 'lock in'
5	11.25
4	9.00
3	9.50

It may be objected that the theory has been of no use in finding the best coil because all the coils had to be wound and tested anyway, in which case the measurements of Table 9 could be made directly to ascertain the optimum. This is so because the coils have properties (loss resistance, mutual inductance) which are difficult to analyse and it is quicker to make the measurements instead. The value of the theory in this example is that it clearly shows why there is an optimal coil and what factors determine it.

5 Conclusions

A steady-state theory has been described for transcutaneous links using series-tuned transmitter and shunt-tuned receiver coils. At the operating frequency given in eqn. 5, some useful properties follow:

- Constant phase angle exists between the voltages across the coils for all k .
- The transmitter tuning is unchanged by any variation in k .
- As k changes, the receiver voltage has a broad maximum at which the coupling efficiency is 50% and at which there is good displacement tolerance.

It is often desirable to have good displacement tolerance and efficiency simultaneously. This will be achieved if the coils can operate at critical coupling without much loss in the receiver coil or the transmitter circuit; as in the first example. If efficiency is of great importance, the coils might be worked at $k > k_{crit}$; conversely, if it is impossible with some coil configurations to obtain k_{crit} low enough, so that $k < k_{crit}$, the efficiency will be poor (Fig. 4); in either case the displacement tolerance will be lower than at critical coupling.

The formulae relating efficiency, voltage transfer ratio and output resistance which have k as the variable can only be converted into spatial values when the variation of coupling coefficient with geometry (coil size, spacing, shape and angular misalignment) is known. Analytical functions for coupling coefficient exist for certain shapes (TERMAN, 1943). For the commonest geometry, two circular coils, we have found that the computed curves of FLACK *et al.* (1971) are in good agreement with our measurements.

In the first example, quantities which are subject to compromise (efficiency, voltage ratio, coil diameter and length) are dependent on things as disparate as battery capacity, threshold for stimulation and what the patient minds wearing. This being so, striking a good balance is not amenable to algebraic optimisation. Instead, working where necessary from measured data, the performance of a design based on reasonable compromises may be calculated and then changes to the design and performance pursued

iteratively until they are satisfactory. This is practicable because of the simplicity of the theoretical formulae used in design.

Generally, decreasing frequency favours efficiency; increasing frequency reduces k_{crit} . Fig. 8 also indicates that series-tuned transmitters are increasingly inefficient beyond about 10 MHz and we have found that correct tuning is then difficult to maintain due to stray capacitance.

In many implants, the voltage is regulated to provide an internal 'power supply' and this means that R_2 is no longer constant. Such a case is considered in the second example. In this case k/k_{crit} is not unity at the coil position under consideration; also, a sufficient number of possible variables have been predetermined for a single coil to be optimal without any compromises being needed.

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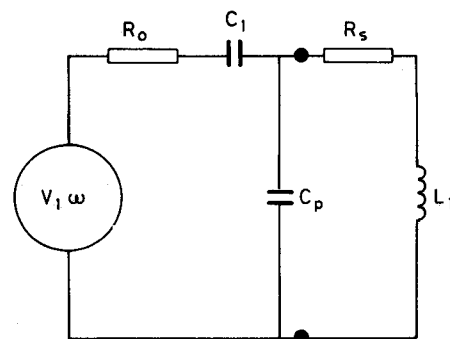


Fig. 15 Transmitter circuit of Fig. 2 but with coil self capacitance represented by C_p in the position shown

TERMAN, F. E. (1943) *Radio Engineers Handbook*. McGraw-Hill Book Co., New York and London.
 VALLEY, G. E. JR. and WALLMAN, H. (1948) *Vacuum tube amplifiers*. McGraw-Hill Book Co., New York, Chap. 5.

Appendix

Correcting for transmitter coil self capacitance

Consider the transmitter circuit of Fig. 1 but with significant self capacitance of the coil appearing as the

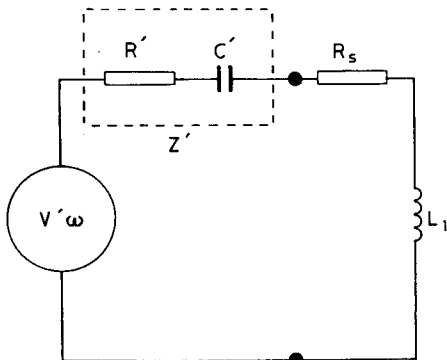


Fig. 16 Thévenin equivalent to the circuit of Fig. 15

capacitor C_p as shown in Fig. 15. To find the effect of this self capacitance, an equivalent of the circuit to the left of the two terminals may be found, as shown in Fig. 16. Applying Thévenin's theorem

$$V' = V_1 \frac{\left(1 + \frac{C_p}{C_1}\right) - j\omega C_p R_0}{\left(1 + \frac{C_p}{C_1}\right)^2 + (\omega C_p R_0)^2} \quad (28)$$

$$Z' = \frac{R_0 - j \left\{ \frac{1}{\omega C_1} \left(1 + \frac{C_p}{C_1}\right) + \omega C_p R_0^2 \right\}}{\left(1 + \frac{C_p}{C_1}\right)^2 + (\omega C_p R_0)^2} \quad (29)$$

For most circuit values of practical interest, these equations may be approximated by

$$C' = C_1 + C_p \quad (30)$$

$$(N.B. \ p = \frac{C_p}{C'} = \frac{C_p}{C_1 + C_p}, \text{ see Section 3.4}).$$

$$V' = V_1(1 - p) \quad (31)$$

$$R' = R_0(1 - p)^2 \quad (32)$$

Generally V' is not in phase with V_1 but the angle $\angle \frac{V'}{V_1}$ will

be constant (i.e. $\angle \frac{V_2}{V_1}$ will be changed but remain independent of k as the coils are brought into proximity). When p is a significant fraction of unity, the quantities V' , R' and C' should be used instead of V_1 , R_0 and C_1 in the formulae of Section 2.

At series resonance in Fig. 16, the total resistance is $R' + R_s$ and this will be doubled by coupled resistance at critical coupling. The power this consumes will actually be supplied by a direct current I_{serit} , and so assuming no losses in the output stage

$$V_s I_{serit} = \frac{V'^2}{2(R' + R_s)} \quad (33)$$

Eliminating V' and R' with eqns. 31 and 32 and assuming V_1 (root mean square) $= \sqrt{2}V_s/\pi$ (see Section 3.6)

$$I_{serit} = \frac{V_s}{\pi^2 \left[R_0 + R_s \left(\frac{1}{1 - p} \right)^2 \right]} \quad (34)$$

This is a theoretical current consumption of the switching output stage due to the coupled resonant circuit at critical coupling. 'Overlap current' and master oscillator current would be added to this.

Although the formulae in this Appendix are useful for making corrections to the theory when self capacitance is significant, the corrections may not be exact because C_p is a fictitious lumped component representing a real distributed capacitance.